

JORGE

ANAIS VILCHEZ

CONTACT

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EDUCATION

Doctorate in Astrophysics and Astroinformatics <i>Universidad de Antofagasta</i> Antofagasta, Chile PhD candidate. Thesis: "Characterization of Sagittarius dwarf galaxy at low Galactic latitudes"	2022 - (2025)
Master in Astronomy <i>Universidad de Antofagasta</i> Antofagasta, Chile Thesis: "Search and characterization of star clusters in the far end of the Galactic bar".	2018 - 2020
Licentiate in Astronomy <i>Pontificia Universidad Católica de Chile</i> Santiago, Chile Graduation activity: "A new astrometric software for the Swope Telescope", developed at Las Campanas Observatory.	2008 - 2014

EXPERIENCE

Early Career Visitor Programme <i>Characterization of new Sagittarius Dwarf Galaxy Clusters</i> ESO, Chile Participated in a project led by Dr. Elisa R. Garro combining deep HST photometry (WFC3, F555W and F814W) and FLAMES+UVES spectroscopy (98 hrs, PI: Garro) to precisely determine ages and chemical compositions of Sagittarius dwarf galaxy globular clusters. Goals include resolving the age-metallicity relation, testing formation scenarios during Sgr-Milky Way interaction, and comparing with theoretical models to reconstruct Sgr's evolutionary history.	04/2025 - Present
Research Project <i>Variable stars in the Sagittarius dwarf spheroidal galaxy</i> NOIRLab Collaboration with NOIRLab researcher Kathy Vivas to detect and classify variable stars (focus on δ Scuti and RR Lyrae stars) in the central regions of Sagittarius galaxy using multiepoch DECam data.	09/2024 - Present
Data Scientist <i>Grupo Kaufmann AWTO</i> Chile and Brasil Design and implement machine learning models, perform exploratory data analysis, communicate insights and recommendations to stakeholders.	08/2021 - 04/2022

SKILLS

Software	Proficient in developing data analysis, reduction, and visualization using Python and its main associated modules (Astropy, Matplotlib, Numpy, Pandas, Plotly, PyTorch, Seaborn, SciPy). Astronomical software: IRAF, DS9, TOPCAT. Other technologies: Bash, Docker, Git, $\text{\LaTeX}$, PostgreSQL, R. Github repo: https://github.com/jorgeanais
Languages	- Spanish (native) - English (professional competence)

TEACHING
EXPERIENCE

Instructor **2022 - Present**
Duoc UC, School of Informatics and Telecommunications | *Santiago, Chile*
Teaching Machine Learning and Deep Learning courses for computer engineering students.

AWARDS

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- 2024-2026 Full Scholarship for advanced human capital in Chile, Agencia Nacional de Investigación y Desarrollo de Chile.
 - 2022-2023 Academic Excellence Scholarship for PhD students, Universidad de Antofagasta.
 - 2019 Full scholarship to attend La Serena School for Data Science, AURA Observatory, Chile.